Chapter 2

METHODOLOGY

This chapter presents the methodology used in the design, development, testing, and evaluation of WorkPulse: Employee Attendance and Work Analytics Information System for MVL. It describes the development model, Scrum activities, deployed system architecture, tools and technologies, respondent population, data gathering, treatment of data, biometric evaluation procedures, data security, and ethical considerations.

WorkPulse centralizes employee, attendance, leave, payroll, cash advance, account, settings, biometric, and audit records. The system provides administrator and employee access groups; employee classifications include Regular, Extra, and Manager / Supervisor. Descriptive reports summarize recorded data, while predictive functions, where implemented and evaluated, provide separate analytical insights for administrative review.

Development Methodology

Software Development Model. The study adopts the Agile Software Development Model using Scrum. Requirements, reported defects, and recommended improvements are prioritized in a product backlog and addressed through iterative sprints. At the start of each sprint, the researchers identify deliverables and acceptance criteria; implemented functions are reviewed and tested, and unfinished tasks return to the backlog.

Figure 1. Phases of Scrum. [Retain the original Scrum figure from the submitted manuscript when integrating this revised text into the final layout.]

Scrum Development Phases

Planning. The researchers identify existing employee-information, attendance, leave, cash advance, payroll, and reporting problems and translate them into functional, security, usability, and deployment requirements. Planning includes fingerprint attendance, administrator manual attendance and corrections, correction history, employee notifications, limited offline saving, payroll settlement protections, authentication, role-based access, and audit logging. Daily and date-range attendance filtering are included in the final-version verification checklist; their completion must be confirmed against the deployed application before being reported as tested.

Design. The researchers design the database collections, user interfaces, authorization boundaries, fingerprint enrollment and attendance procedures, leave and cash advance requests, payroll workflows, offline queue, synchronization process, and analytical reporting. The employee-management module maintains employee profiles, classifications, assignments, and other authorized employment records. The payroll module manages authorized pay inputs, payroll computation and review, cash advance deductions, carried balances, settlement, and employee payslip summaries. Payroll settlement prevents unauthorized retroactive attendance changes affecting finalized payroll.

Development. The React frontend implements the administrator workspace, employee portal, and attendance kiosk. The Node.js and Express.js backend implements protected APIs, business rules, validation, attendance processing, leave and cash advance workflows, payroll processing, analytics, notifications, and audit logging. MongoDB Atlas stores authorized cloud records. IndexedDB and service worker caching support specifically designated offline actions and previously cached content.

Testing. Functional tests cover login, account ownership, employee management, fingerprint enrollment and attendance, manual attendance and corrections, leave, cash advances, payroll, reports, notifications, offline queue, and synchronization. Boundary and negative tests cover future-dated attendance times, overlapping attendance sessions, attempts to edit attendance already included in settled payroll, unauthorized access, stale cached records, duplicate synchronization submissions, network interruptions, and local-storage failures. For each test, the researchers record the input, expected result, actual result, and disposition; the manuscript must not imply a test passed unless evidence exists.

Deployment. The frontend is hosted on Vercel and accessed through a web browser. The frontend sends authenticated HTTPS requests to the Node.js/Express.js backend hosted on Render. The Render backend enforces server-side authorization and business rules and reads or writes cloud records in MongoDB Atlas. A configured local fingerprint workstation hosts the compatible scanner, reader interface, and any locally required template extraction or matching components. Scanner access is local to that workstation; a remote Render service cannot directly read USB scanner hardware or execute software installed only on the workstation.

Limited Offline Operation. IndexedDB stores permitted queued actions and selected cached records in the browser; service worker caching supports designated application resources.

Subject to final implementation verification, offline users may queue authorized manual attendance/corrections, leave requests, and cash advance requests and view previously cached information. Cached records may be outdated. When connectivity returns, queued items are synchronized to the server, revalidated against current permissions and rules, and marked Synced or Needs Review. Fingerprint scanning, payments, approvals, fresh login, and operations requiring live server verification are not available offline. WorkPulse is not a fully offline system.

System Architecture Overview

WorkPulse uses a multi-layered architecture connecting browser-based presentation, server-side application services, cloud data storage, and locally operated fingerprint components. Fingerprint capture and verification replace the unrelated facial-recognition description in the earlier manuscript.

Presentation Tier. React 19, TypeScript, Vite, Tailwind CSS, and Recharts provide administrator, employee, and kiosk interfaces. IndexedDB supports browser-local queued records, and a service worker caches eligible application assets. User interface visibility is not a substitute for backend authorization.

Application Tier. Node.js and Express.js on Render provide authenticated REST endpoints, enforce role-based access and record ownership, validate requests, prevent duplicate synchronization, protect settled payroll, and handle payroll, attendance, leave, cash advance, notification, and audit workflows.

Data Tier. MongoDB Atlas stores authorized employee, account, attendance, leave, cash advance, payroll, settings, biometric-related, session, and audit records. Browser-local data is temporary and must be reconciled with authoritative server records.

Local Biometric Tier. The configured fingerprint workstation connects to the DigitalPersona reader through locally installed reader components and the applicable WebSDK or bridge. FingerJet template extraction or matching runs only where the required native runtime and files are actually installed and accessible. If FingerJet is installed only on the local workstation, the local workstation performs that step and communicates the resulting permitted data or decision securely to the backend. The exact deployed processing location and trust boundary must be checked against the working source code and workstation setup.

Email Integration. The backend may support SMTP through Nodemailer or Brevo HTTPS delivery. The final manuscript must identify the provider and delivery method actually configured in the deployed environment; support for both methods in code does not prove that both were used in production.

Figure 2. Revised WorkPulse Deployment and Data Flow

Web browser (Administrator / Employee / Kiosk) — React frontend hosted on Vercel

↳ IndexedDB offline queue + service worker cache (browser-local)

↕ HTTPS authenticated API requests / synchronization

Node.js + Express.js backend hosted on Render

↕ authorized database operations

MongoDB Atlas (authoritative cloud records)

↘️ configured email delivery (SMTP/Nodemailer or Brevo HTTPS)

Local fingerprint workstation → DigitalPersona reader / local capture components → locally configured FingerJet processing → protected application workflow

Figure note: The local fingerprint workstation is not a USB device attached to Render. This schematic should be reconciled with the final deployment and substituted for the original Figure 2 in the typeset manuscript.

Tools and Technologies

React 19, TypeScript, Vite, Tailwind CSS, and Recharts. Provide the responsive web interfaces, typed frontend development, bundling, styling, and graphical reporting.

Node.js and Express.js. Provide the backend runtime, protected REST API routes, application services, and server-side validation. Node.js native scrypt with randomly generated salts is used for password derivation where implemented.

MongoDB Atlas and Mongoose/MongoDB operations. Provide cloud-hosted record storage and database access.

IndexedDB. Stores permitted browser-local cached records and pending offline submissions, subject to browser storage limitations and account separation.

Service Worker. Caches selected application assets to support limited offline use; it does not replace a server or make every feature available without connectivity.

Vercel. Hosts and delivers the deployed React frontend.

Render. Hosts the deployed Node.js and Express.js backend.

DigitalPersona fingerprint library/WebSDK and compatible scanner. Support fingerprint capture at the configured local workstation.

HID FingerJet. Supports fingerprint-template processing and comparison in the runtime environment where it is actually installed.

Email services. Nodemailer/SMTP and Brevo HTTPS are supported delivery options; the actually configured production provider requires confirmation.

Dotenv, Git, and GitHub. Support configuration and, where actually used, source-code version control and collaboration. Any claim of biometric template encryption or a specific cryptographic algorithm should be retained only if verified in the final code and configuration.

Population and Locale of the Study

The study uses total enumeration of the 30 identified MVL respondents: two owners, two supervisors/managers, ten regular employees, and sixteen extra employees. These are organizational respondent categories rather than four separate WorkPulse login roles. The system uses administrator and employee access groups; Manager / Supervisor, Regular, and Extra are employee classifications. The two owners are counted as respondents and may perform administrator tasks according to their assigned access. The study does not add respondents when describing access groups.

The study locale is the MVL workplace in Salay Riles, Mangaldan, Pangasinan, Region I. Authorized actual records and suitable simulated records may be used during testing to limit exposure of personal information.

Table 1. Respondents of the Study

Respondent Classification	Number of Respondents
Owner	2
Supervisor / Manager	2

Regular Employees	10
Extra Employees	16
Total	30

Data Gathering Techniques

Interviews. Semi-structured interviews with authorized MVL owners and supervisors/managers identify actual attendance, employee-information, leave, cash advance, payroll, policy, reporting, and operational requirements. A supervisor or manager participating as an employee uses the Manager / Supervisor classification rather than being described as a regular employee.

Survey Questionnaires. Eligible respondents answer the applicable evaluation questionnaire after performing authorized WorkPulse tasks. The questionnaire addresses selected software-quality criteria such as functional suitability, usability, performance efficiency, reliability, and perceived security. Claims that three experts validated the instrument, a separate pilot test was conducted, or Cronbach's alpha was computed require actual supporting documents and data. Without such evidence, these activities must be described as planned or omitted; no coefficient is asserted here.

Observation. Researchers document the existing MVL processes for attendance capture, attendance correction, employee records, leave, cash advances, payroll, and payslip distribution. Observations are used to identify bottlenecks and compare the implemented workflows with actual workplace practices.

Sources of Data

Primary data are obtained from the 2 owners, 2 supervisors/managers, 10 regular employees, and 16 extra employees who participate in interviews, observations, task-based testing, or questionnaires as applicable. Owners and supervisors/managers provide policy and approval requirements; regular and extra employees provide information on attendance and employee self-service experiences.

The researchers observe MVL workplace recordkeeping, attendance logging, leave processing, cash advance handling, payroll preparation, and report generation, subject to permission and privacy safeguards. These observations replace the unrelated school tracing and alumni

information discussion. Secondary data may include relevant technical references, system documentation, and research literature.

Biometric Evaluation Procedures

Participants and Consent. Invite authorized consenting participants who meet the study eligibility criteria. Record the actual number of enrolled participants and scan trials in the evaluation log; do not invent sample sizes. Explain the purpose of fingerprint testing, privacy measures, and the approved alternative attendance procedure.

Enrollment. At the configured workstation, enroll the designated fingerprint according to the reader and matching-software procedure. Record successful enrollments, capture failures, repeated attempts, and any exclusions. Separate failed captures from valid comparisons.

Genuine Identification Trials. An enrolled participant presents the finger enrolled in the evaluation database. The system searches enrolled templates. A correct identification is a genuine success; a failure to identify the enrolled participant is a false rejection when a valid comparison occurred.

Unrecognized / Impostor Trials. A person or finger not enrolled in the evaluation database is presented for one-to-many identification. The expected result is no match. Any match to an enrolled employee is a false acceptance. An enrolled participant presenting their enrolled finger is not an impostor trial.

Conditions and Timing. Document the scanner and software versions, workstation setup, connection status, finger condition where relevant, trial sequence, capture time, template processing time, matching time, and total response time. Report actual measurements and capture failures separately.

Outcome Calculation. For valid impostor comparisons, False Acceptance Rate (FAR) = false acceptances ÷ valid impostor comparisons × 100%. For valid genuine comparisons, False Rejection Rate (FRR) = false rejections ÷ valid genuine comparisons × 100%. Report the count of failed captures separately and state the denominator used. Actual values belong in the results chapter after testing.

Treatment of Data

Data Collection and Processing. Interview notes, survey responses, observation records, and testing logs are checked for completeness. Authorized datasets are organized consistently, with date and time formats standardized and duplicates identified without changing participants' original responses.

Data Analysis. Descriptive summaries, frequencies, percentages, weighted means, and, where applicable, usability scores describe respondents' evaluation and workforce records. Descriptive dashboards summarize historical information; predictive outputs are a separate function and should be reported only for models actually implemented and evaluated. Any trained model must be distinguished from statistical fallback outputs.

Data Storage and Backup. MongoDB Atlas stores authoritative cloud records subject to authentication, authorization, ownership restrictions, and server-side validation. IndexedDB temporarily stores permitted account-separated browser data and queued actions. Pending entries await submission, Synced entries have been accepted by the server, and Needs Review entries require user or administrator action after validation or conflict detection. Synchronization must recheck current access, attendance rules, payroll settlement, and record versions and prevent duplicate processing through unique operation identifiers or equivalent idempotency controls. Cached information may be stale, and browser storage can be cleared or fail. Backup and restoration claims must reflect the configured deployment and actual tests; cloud storage alone does not establish that a recovery test occurred.

Ethical Considerations

Security and Privacy of Data. WorkPulse handles employee records, attendance, leave, cash advances, payroll, and fingerprint biometric data/templates. Data processing should follow Republic Act No. 10173 and principles of transparency, legitimate purpose, proportionality, accountability, and reasonable safeguards. Only authorized processes should access biometric templates and sensitive workforce records.

Fair and Inclusive Design. Fingerprint attendance verifies identity; it is not a direct measure of employee performance. Employees who cannot complete fingerprint capture may use verified administrator manual attendance under an approved policy. The administrator records the reason, affected employee, time, and correction history, and the employee receives a notification when supported. Attendance included in settled payroll must not be silently edited;

changes require the designated review or adjustment process. If the local workstation is unavailable, a controlled paper attendance log provides a fallback for later authorized entry and reconciliation.

Transparency and Consent. Participants are informed of the data collected, the purpose of fingerprint enrollment and verification, who may review records, and how testing data are handled. Participation in research evaluation is voluntary, and the study must not imply that choosing a proposed biometric option constitutes enrollment or organizational approval. Analytical outputs support human review rather than automatic disciplinary or employment decisions.

Editorial verification note (remove before submission): Confirm final date-range filtering, exact FingerJet execution location, deployed email provider, browser offline behavior, biometric encryption implementation, questionnaire validation/pilot records, biometric trial logs, and backup restoration evidence. No unverified results have been invented in this revision.